

2.4 Course Content

UNSTRUCTURED AND SEMI-STRUCTURED DATA

Limitations of Relational Databases: Fixed schema constraints, impedance mismatch, difficulty in handling hierarchical and evolving data, scalability challenges.

Types of Data: Structured data, Semi-structured data, Unstructured data.

Characteristics of Unstructured Data: Schema flexibility, Hierarchical and nested data, Heterogeneous attributes, Rapid schema evolution.

Data Representation Formats: JSON structure and syntax, BSON representation, XML (overview and comparison), Differences between JSON and relational tables.

Use Cases of Unstructured Data: Web applications, Content management systems, Social media data, Logs and event data, IoT-generated data.

INTRODUCTION TO NoSQL DATABASES

Need for NoSQL Databases: Scalability requirements, High availability, Distributed data storage, Handling large volumes of unstructured data.

Overview of NoSQL Database Models: Document-oriented databases, Key-Value stores, Column-family databases, Graph databases.

CAP Theorem: Consistency, Availability, Partition tolerance, Trade-offs in distributed systems (conceptual understanding).

Comparison of SQL vs NoSQL Databases: Data model differences, Query mechanisms Schema enforcement, Transaction support, Typical application scenarios.

DOCUMENT-ORIENTED DATABASES

Document Data Model: Documents and collections, Nested documents and arrays, Schema-less vs schema-flexible design.

Introduction to MongoDB: MongoDB architecture (logical view), Databases, collections, and documents, BSON data types.

CRUD Operations in MongoDB: Insert operations, Query operations with filters, Update operations, Delete operations.

Querying Documents: Projection, Conditional queries, Sorting and limiting results, Pattern matching.

Indexing in MongoDB: Purpose of indexing, Single-field and compound indexes, Impact of indexing on query performance.

Aggregation Framework: Aggregation pipeline concept, Stages such as match, group, project, sort, Data summarization and analytics.

Data Modeling in MongoDB: Embedding vs referencing, One-to-one, one-to-many, and many-to-many relationships, Modeling patterns for real-world applications transactions and

Consistency: Atomic operations, Multi-document transactions (introductory level).

Practical Orientation: Extensive hands-on development using MongoDB for data modeling, querying, indexing, and aggregation.

GRAPH DATABASES AND GRAPH DATA MODELS

Introduction to Graph Data: Nodes, relationships (edges), and properties, Directed and undirected graphs, Labeled property graph model.

Why Graph Databases: Relationship-centric data, Traversal-based queries, Comparison with relational and document databases.

Graph Database Concepts: Nodes and relationships, Properties on nodes and edges, Path traversal.

Graph Querying: Pattern matching concepts, Traversals and shortest paths (conceptual), Use of graph query languages (introductory).

Applications of Graph Databases: Social networks, Recommendation systems, Fraud detection, Knowledge graphs.

Practical Orientation: Introductory exposure through simple graph creation and basic traversal queries using a graph database tool.

OTHER NoSQL MODELS AND ADVANCED CONCEPTS

Key–Value Databases: Key–value data model, Basic operations, Typical use cases such as caching and session storage.

Column-Family Databases: Column-oriented storage, Row keys and column families, High-level use cases.

Overview of Popular NoSQL Systems: MongoDB (document), Neo4j (graph), Redis (key–value), Cassandra (column-family).

NoSQL in Modern Applications: Integration with web frameworks, Microservices and NoSQL, Polyglot persistence.

Data Security and Backup (Introductory): Authentication and authorization concepts, Backup and restore basics.